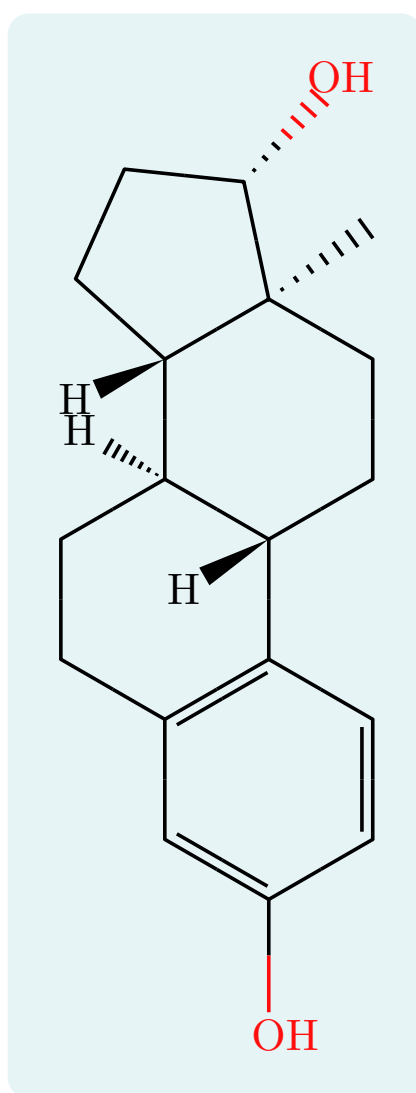


# typed-smiles

Render SMILES strings as 2D molecular diagrams

## User Guide



Version 0.5.0

# Contents

---

|      |                                                                                        |    |
|------|----------------------------------------------------------------------------------------|----|
| 1    | Introduction .....                                                                     | 3  |
| 2    | Getting started .....                                                                  | 4  |
| 2.1  | Installation and import .....                                                          | 4  |
| 2.2  | Your first molecule .....                                                              | 4  |
| 3    | The <code>smiles()</code> function .....                                               | 5  |
| 3.1  | Argument reference .....                                                               | 5  |
| 3.2  | <code>smiles-str</code> .....                                                          | 7  |
| 3.3  | <code>style</code> .....                                                               | 8  |
| 3.4  | <code>scale</code> .....                                                               | 9  |
| 3.5  | <code>bond-length</code> .....                                                         | 9  |
| 3.6  | <code>font-size</code> .....                                                           | 9  |
| 3.7  | <code>font</code> .....                                                                | 10 |
| 3.8  | <code>bond-stroke</code> .....                                                         | 10 |
| 3.9  | <code>color</code> .....                                                               | 11 |
| 3.10 | <code>fg</code> and <code>theme</code> .....                                           | 11 |
| 3.11 | <code>rotation</code> .....                                                            | 12 |
| 3.12 | <code>mirror</code> .....                                                              | 13 |
| 3.13 | <code>show-h</code> .....                                                              | 15 |
| 3.14 | <code>atom-annotations</code> .....                                                    | 16 |
| 3.15 | <code>opacity</code> .....                                                             | 16 |
| 3.16 | <code>bond-customizations</code> .....                                                 | 17 |
| 3.17 | <code>lone-pairs</code> .....                                                          | 17 |
| 4    | Atoms and charges .....                                                                | 18 |
| 4.1  | Aromatic rings .....                                                                   | 18 |
| 4.2  | Charges .....                                                                          | 21 |
| 4.3  | Salts and disconnected structures .....                                                | 21 |
| 5    | Hydrogens .....                                                                        | 22 |
| 5.1  | Default display .....                                                                  | 22 |
| 5.2  | Explicit bracket hydrogens .....                                                       | 22 |
| 5.3  | Isotopes .....                                                                         | 22 |
| 5.4  | Label orientation .....                                                                | 23 |
| 6    | Stereochemistry .....                                                                  | 24 |
| 6.1  | Tetrahedral centers: <code>@</code> and <code>@@</code> .....                          | 24 |
| 6.2  | Double bonds: <code>/</code> and <code>\</code> .....                                  | 25 |
| 6.3  | Square-planar centers: <code>@SP1</code> , <code>@SP2</code> , <code>@SP3</code> ..... | 25 |
| 6.4  | Manual wedges: <code>!w</code> and <code>!h</code> .....                               | 26 |
| 6.5  | Wavy and dashed bonds: <code>!s</code> and <code>!d</code> .....                       | 26 |
| 7    | Colors .....                                                                           | 28 |
| 7.1  | Jmol CPK palette .....                                                                 | 28 |
| 7.2  | Custom colors ( <code>atom-colors</code> ) .....                                       | 28 |
| 7.3  | Label colors ( <code>{label style}</code> ) .....                                      | 30 |
| 8    | Abbreviated groups .....                                                               | 32 |
| 8.1  | Plain labels .....                                                                     | 32 |
| 8.2  | Colored labels .....                                                                   | 32 |
| 9    | Chemical formulas: <code>ce()</code> .....                                             | 33 |
| 9.1  | Fonts and size .....                                                                   | 33 |
| 10   | Molecular weight: <code>mol-weight()</code> .....                                      | 34 |
| 11   | Inline molecules: <code>smiles-inline()</code> .....                                   | 35 |

|      |                                                         |    |
|------|---------------------------------------------------------|----|
| 12   | CeTZ integration: <code>smiles-cetz()</code> .....      | 36 |
| 13   | Reaction schemes .....                                  | 38 |
| 13.1 | <code>mol()</code> .....                                | 38 |
| 13.2 | <code>rxn-arrow()</code> .....                          | 38 |
| 13.3 | <code>reaction()</code> .....                           | 40 |
| 14   | Reaction mechanisms .....                               | 45 |
| 14.1 | Referencing atoms .....                                 | 45 |
| 14.2 | <code>arrow()</code> and <code>highlight()</code> ..... | 46 |
| 14.3 | A mechanism across species .....                        | 48 |
| 14.4 | <code>brackets()</code> .....                           | 48 |
| 15   | Catalytic cycles: <code>cycle()</code> .....            | 49 |
| 16   | Project-wide defaults .....                             | 52 |
| 17   | Limitations .....                                       | 53 |
| 18   | Quick reference .....                                   | 54 |
| 18.1 | Syntax .....                                            | 54 |
| 18.2 | <code>smiles()</code> options .....                     | 54 |
| 18.3 | <code>reaction()</code> options .....                   | 55 |
| 18.4 | <code>rxn-arrow()</code> options .....                  | 55 |
| 18.5 | Data helpers .....                                      | 55 |
| 18.6 | Mechanism helpers .....                                 | 56 |
| 18.7 | Label color names .....                                 | 57 |

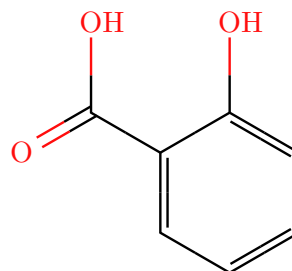
# 1 Introduction

typed-smiles renders SMILES strings as 2D skeletal molecular diagrams inside Typst documents. A bundled WebAssembly plugin parses the SMILES, computes atom coordinates, implicit hydrogens, and stereochemistry, and the Typst layer draws the bonds, atom labels, colors, charges, hydrogens, wedges, and reaction helpers.

This guide documents every function, argument, and package-specific syntax extension. Each feature comes with a runnable example: the source is on the left, its rendered output on the right.

```
1 #smiles("OC1=CC=CC=C1C(=O)O")
```

 Typst



## 2 Getting started

### 2.1 Installation and import

Import the package from the Typst preview namespace. A wildcard import gives you every public symbol:

```
1 #import "@preview/typed-smiles:0.5.0": *
```

 Typst

Or import only what you need:

```
1 #import "@preview/typed-smiles:0.5.0": smiles, ce, mol, rxn-arrow, reaction
```

 Typst

The package exports these main symbols:

| Symbol                  | Purpose                                                                      |
|-------------------------|------------------------------------------------------------------------------|
| smiles                  | Render a SMILES string as a molecule (the main function).                    |
| ce                      | Chemical formulas and equations, re-exported from <code>chemformula</code> . |
| mol                     | Wrap a molecule with an optional caption.                                    |
| rxn-arrow               | A reaction arrow with conditions above and below.                            |
| reaction                | Lay out a multi-step reaction scheme.                                        |
| atom, bond, lp, species | References for arrows and highlights.                                        |
| arrow, highlight        | Mechanism arrows and shaded atom/bond regions.                               |
| brackets                | Draw square brackets with optional corner marks.                             |
| mol-weight              | Compute molecular weight from a SMILES string.                               |

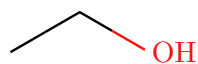
### 2.2 Your first molecule

Call `smiles` with a SMILES string.

```
1 Ethanol: #smiles("CCO")
```

 Typst

Ethanol:



## 3 The smiles() function

smiles is the main function. Its full signature is:

```
1  #smiles(  
2    smiles-str,  
3    style: "default",  
4    scale: 1.0,  
5    bond-length: none,  
6    font-size: none,  
7    font: auto,  
8    bond-stroke: none,  
9    color: auto,  
10   fg: auto,  
11   theme: auto,  
12   rotation: 0deg,  
13   mirror: none,  
14   show-h: (),  
15   aromatic: "kekule",  
16   atom-annotations: (),  
17   opacity: 100%,  
18   bond-customizations: (),  
19   lone-pairs: none,  
20   atom-colors: (:),  
21   show-indices: false,  
22 )
```

 Typst

### 3.1 Argument reference

| Argument    | Type          | Default    | Description                                                                                                                                                     |
|-------------|---------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| smiles-str  | str           | (required) | The SMILES string.                                                                                                                                              |
| style       | str           | "default"  | Journal style preset ("acs", "rsc", "nature", "wiley") filling in bond length, label size, stroke, and font. Explicit arguments win; "default" applies nothing. |
| scale       | float         | 1.0        | Balanced scale for bond length, label size, and stroke at once.                                                                                                 |
| bond-length | float / none  | none       | Bond length factor (1.0 is 30 pt). Overrides scale for length.                                                                                                  |
| font-size   | length / none | none       | Atom-label font size. Overrides scale for labels.                                                                                                               |

|                     |                                  |          |                                                                                                                               |
|---------------------|----------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------|
| font                | auto / str / array               | auto     | Font family for atom labels; <b>auto</b> is “New Computer Modern”, or the preset’s font when <b>style</b> is set.             |
| bond-stroke         | length / none                    | none     | Bond stroke width. Overrides <b>scale</b> for strokes.                                                                        |
| color               | auto / bool                      | auto     | Apply Jmol CPK atom colors. <b>auto</b> is colored for “default” and monochrome for journal presets.                          |
| fg                  | auto / color                     | auto     | Foreground for bonds and carbon labels; <b>auto</b> inherits the surrounding text color.                                      |
| theme               | auto / “light” / “dark”          | auto     | CPK palette variant; <b>auto</b> picks “dark” when <b>fg</b> is light.                                                        |
| rotation            | angle                            | 0deg     | Rotate the molecule; labels stay upright.                                                                                     |
| mirror              | none / “horizontal” / “vertical” | none     | Optional horizontal or vertical page-axis reflection.                                                                         |
| show-h              | “all” / int / array              | ()       | Implicit hydrogens to label beyond the default heteroatom hydrogens. Use “all” for every atom.                                |
| aromatic            | “kekule” / “circle”              | “kekule” | Depict lowercase-aromatic rings with alternating double bonds or with an inscribed circle.                                    |
| atom-annotations    | array                            | ()       | Small gray side labels as (index, content) or (index, content, offset) tuple entries.                                         |
| opacity             | ratio / float                    | 100%     | Fade the whole drawing — bonds, labels, charges, lone pairs — e.g. for ghost molecules.                                       |
| bond-customizations | array                            | ()       | Per-bond style overrides as (bond(i, j), (.options..)) pairs; options are <b>color</b> , <b>stroke</b> , and <b>opacity</b> . |

|             |                         |      |                                                                                                                                                                                                                                                |
|-------------|-------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lone-pairs  | none / "dots" / "lines" | none | Draw optional non-bonding electron pairs on skeletal atom labels.                                                                                                                                                                              |
| atom-colors | dictionary              | (:)  | Color overrides for elements and labels. Element-symbol keys (e.g. <code>O:red</code> ) override CPK colors; brace-quoted label keys (e.g. <code>{PPh3}:purple</code> ) override a specific abbreviated group. See <a href="#">Section 7</a> . |

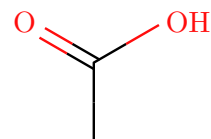
**Note.** `scale` sizes everything together. Use `bond-length`, `font-size`, and `bond-stroke` to override one dimension on its own; each defaults to `none`, meaning it follows `scale`.

## 3.2 smiles-str

The positional argument is the SMILES string.

```
1 #smiles("CC(=O)O")
```

 Typst





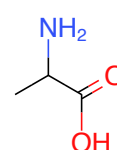
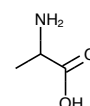
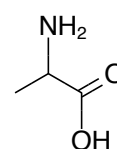
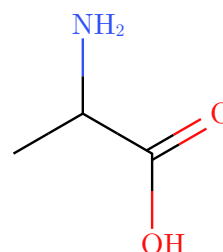
### 3.3 style

Journal style presets approximate the corresponding ChemDraw document stylesheets using the journals' published drawing settings — bond length, atom-label size, line width, a Helvetica/Arial font stack, and monochrome atom labels by default:

| Preset   | Bond    | Line   | Labels |
|----------|---------|--------|--------|
| "acs"    | 14.4 pt | 0.6 pt | 10 pt  |
| "rsc"    | 12.2 pt | 0.5 pt | 7 pt   |
| "nature" | 10.8 pt | 0.6 pt | 6 pt   |
| "wiley"  | 17 pt   | 0.7 pt | 12 pt  |

```
1 #smiles("CC(N)C(=O)O") \  
2 #smiles("CC(N)C(=O)O", style: "acs") \  
3 #smiles("CC(N)C(=O)O", style: "nature") \  
4 #smiles("CC(N)C(=O)O", style: "acs",  
  color: true)
```

 Typst

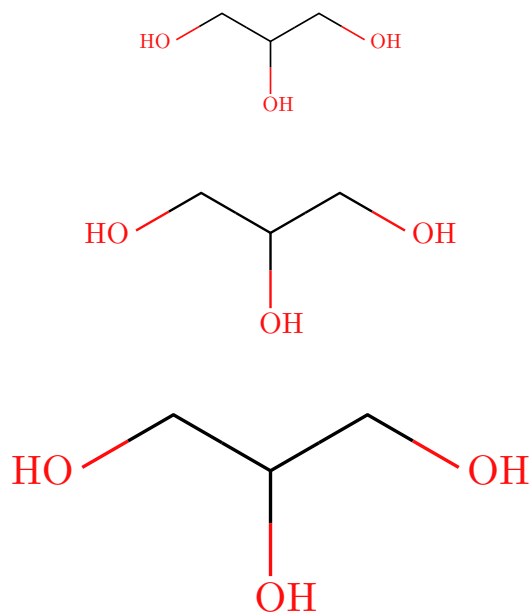


A preset only fills in arguments left unset: `style: "acs"` with `font-size: 14pt` keeps the ACS bond length under your label size, and `scale` multiplies the whole preset. `color: auto` makes journal presets monochrome; pass `color: true` for CPK colors or `color: false` for explicit black line art. `default` applies nothing, so existing documents are unaffected. Wiley's line width and label size are derived from its 6 mm bond-length guideline, which fixes only a minimum line width; journal-specific geometry beyond these four settings (double-bond spacing, hash spacing) keeps typed-smiles' defaults.

### 3.4 scale

Scales bond length, labels, and stroke proportionally.

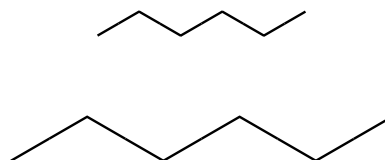
```
1 #smiles("OCC(O)CO", scale: 0.7) \  Typst
2 #smiles("OCC(O)CO", scale: 1.0) \
3 #smiles("OCC(O)CO", scale: 1.4)
```



### 3.5 bond-length

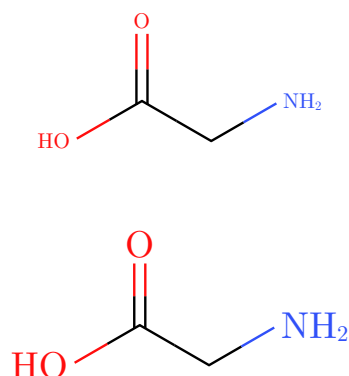
Sets bond length on its own (1.0 is 30 pt per bond), leaving label size and stroke under scale.

```
1 #smiles("CCCCC", bond-length:  Typst
  0.6) \
2 #smiles("CCCCC", bond-length: 1.1)
```



### 3.6 font-size

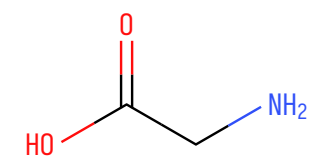
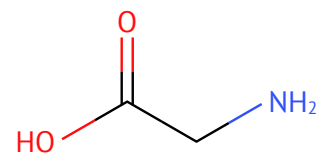
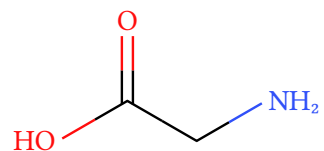
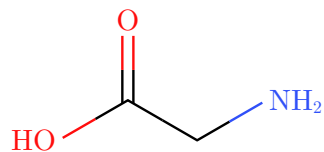
```
1 #smiles("OC(=O)CN", font-size:  Typst
  8pt) \
2 #smiles("OC(=O)CN", font-size: 14pt)
```



### 3.7 font

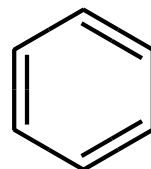
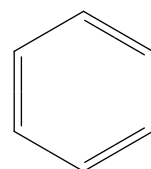
Match the document typeface, or pick something with character.

```
1 #smiles("OC(=O)CN", font: "New  
  Computer Modern") \  Typst  
2 #smiles("OC(=O)CN", font: "Libertinus  
  Serif") \  
3 #smiles("OC(=O)CN", font: "PT Sans") \  
4 #smiles("OC(=O)CN", font: "Iosevka")
```



### 3.8 bond-stroke

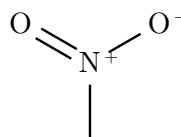
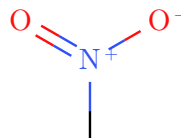
```
1 #smiles("C1=CC=CC=C1", bond-  
  stroke: 0.5pt) \  Typst  
2 #smiles("C1=CC=CC=C1", bond-stroke: 1.6pt)
```



### 3.9 color

CPK colors are on by default for `style: "default"` (see [Section 7](#)). Journal presets default to monochrome line art. Set `color: false` for an explicit monochrome diagram, or `color: true` to force CPK colors.

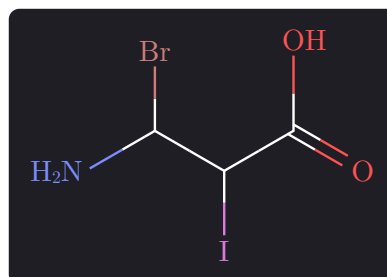
```
1 #smiles("C[N+](=O)[O-]") \
2 #smiles("C[N+](=O)[O-]", color: false)
```

 Typst

### 3.10 fg and theme

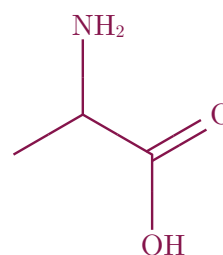
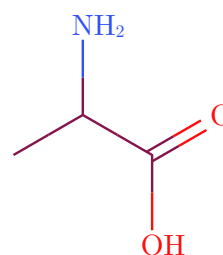
Bond strokes and carbon labels follow `fg`, which defaults to `auto` and inherits the surrounding text color — on a dark slide theme (Touying, Polylux, or any `set text(fill: white)`) molecules recolor with no configuration. `theme: auto` switches to a dark CPK variant whenever the foreground is light, lifting only the hues that vanish on dark backgrounds (N, O, Br, I, and dark named label colors such as navy or maroon).

```
1 #block(fill: rgb("#1E1E24"),
  inset: 8pt,
2   radius: 4pt)[
3   #set text(fill: white)
4   #smiles("NC(Br)C(I)C(=O)O")
5 ]
```

 Typst

Both can be forced: an explicit `fg` recolors the skeleton on any background, and `theme: "dark"` or `theme: "light"` pins the palette. To set a document-wide default, rebind once in the preamble: `#let smiles = smiles.with(fg: white, theme: "dark")`.

```
1 #smiles("CC(N)C(=O)O", fg:
  maroon) \
2 #smiles("CC(N)C(=O)O", color: false, fg:
  maroon)
```

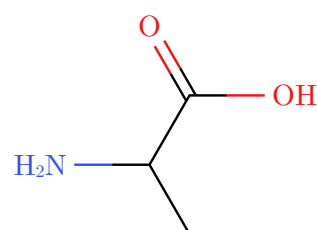
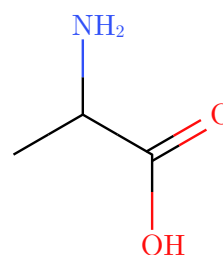
 Typst

**Note.** Reaction arrows inherit the surrounding text color the same way via `rxn-arrow(color: auto)`, so full schemes work on dark slides too.

### 3.11 rotation

Rotates the whole molecule while keeping every label upright.

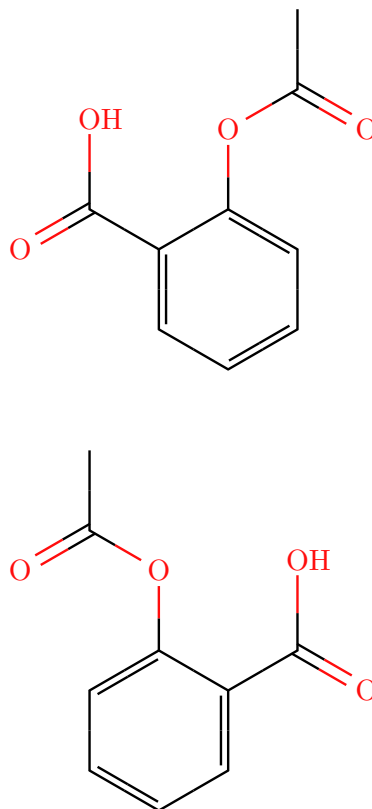
```
1 #smiles("CC(N)C(=O)O", rotation:
  0deg) \
2 #smiles("CC(N)C(=O)O", rotation: 90deg)
```

 Typst

### 3.12 mirror

`mirror` reflects a molecule horizontally or vertically in the final drawing. Useful to face a reacting group toward a reaction arrow. Inside `reaction()`, pass it per molecule: `mol("OC1CCCCC1", mirror: "horizontal")`. With `rotation` set, `mirror: "vertical"` preserves left/right and flips top/bottom.

```
1 #smiles("CC(=O)OC1=CC=CC=C1C(=O)O")  Typst
2 \
3 #smiles(
4   "CC(=O)OC1=CC=CC=C1C(=O)O",
5   mirror: "horizontal",
6 )
```

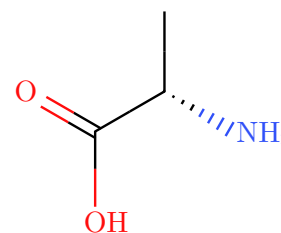
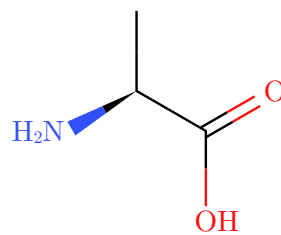


**Note.** Single-axis reflection exchanges wedges and hashes: a reflected 2D skeleton with unchanged wedges would depict the enantiomer, while reflecting and swapping is equivalent to turning the molecule over — the depicted configuration at every stereocenter is preserved.

A stereocenter keeps its configuration when mirrored.

```
1 #smiles("N[C@@H](C)C(=O)O") \  
2 #smiles(  
3   "N[C@@H](C)C(=O)O",  
4   mirror: "horizontal",  
5   )
```

 Typst

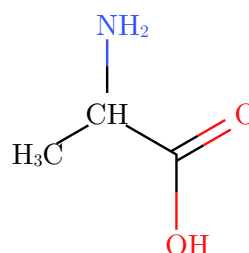
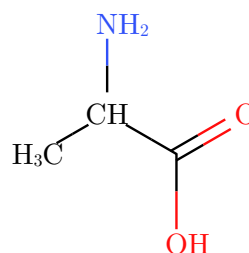
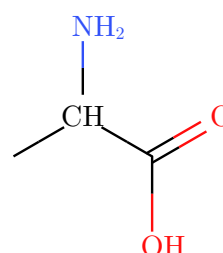
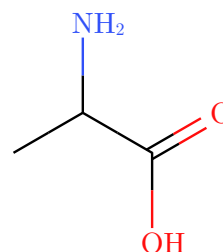


### 3.13 show-h

Hydrogens on heteroatoms are shown by default; carbon hydrogens are hidden. `show-h` selects carbon hydrogens without needing a trailing comma for one atom: use an integer for one atom, an array for several atoms, or "all" for every implicit hydrogen. Use `show-indices: true` to read off the indices while writing.

```
1 #smiles("CC(N)C(=O)O") \  
2 #smiles("CC(N)C(=O)O", show-h: 1) \  
3 #smiles("CC(N)C(=O)O", show-h: (0, 1)) \  
4 #smiles("CC(N)C(=O)O", show-h: "all")
```

 Typst



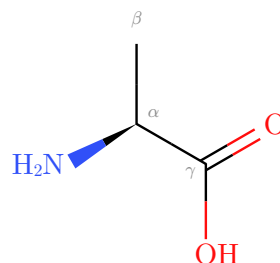


### 3.14 atom-annotations

Small annotations for NMR numbering, Greek positions, or footnote marks are drawn small and gray on the emptiest side of an atom. `atom-annotations` is always a tuple list. Each entry is either `(index, content)` or `(index, content, offset)`. Values are content, so wrap them in `text()` to restyle.

```
1 #smiles(  
2   "N[C@@H](C)C(=O)O",  
3   atom-annotations: (  
4     (1, [$alpha$], (0, -0.05)),  
5     (2, [$beta$]),  
6     (3, [$gamma$], (-0.4, -0.3)),  
7   ),  
8 )
```

 Typst

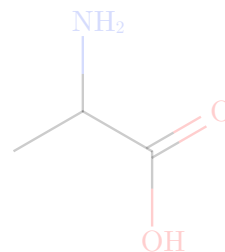
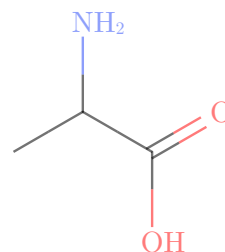
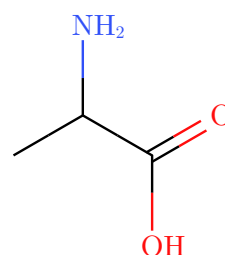


### 3.15 opacity

`opacity` fades every paint the drawing produces — bonds, atom labels, charges, lone pairs, and annotations — so a molecule can be ghosted or de-emphasized, e.g. spectator species in a mechanism or step-by-step slide builds. It accepts a ratio (40%) or a number between 0 and 1.

```
1 #smiles("CC(N)C(=O)O")  
2 #h(1.5em)  
3 #smiles("CC(N)C(=O)O", opacity: 55%)  
4 #h(1.5em)  
5 #smiles("CC(N)C(=O)O", opacity: 0.2)
```

 Typst



**Note.** String molecules inside `reaction()` accept the same option: `mol("CCO", opacity: 30%)`.

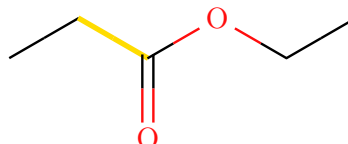
### 3.16 bond-customizations

`bond-customizations` restyles individual bonds. Each entry pairs a `bond(i, j)` reference (the same atom-index form used by `arrow()` and `highlight()` a plain `(i, j)` array also works) with an options dictionary:

- `color` — draw the whole bond in one color instead of the bicolor halves,
- `stroke` — the bond width, like a per-bond `bond-stroke`,
- `opacity` — fade just this bond.

Overrides apply to every part of the bond: both lines of a double bond, the hash lines of a stereo bond, waves and dashes.

```
1 #smiles(  
2   "CCC(=O)OCC",  
3   bond-customizations: (  
4     (bond(1, 2), (color: yellow, stroke:  
5       2pt)),  
6   ),  
7 )
```

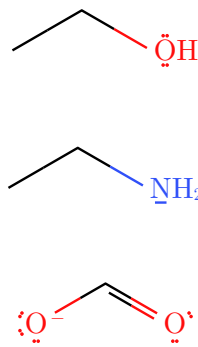
 Typst

**Note.** Use `show-indices: true` while writing the references, exactly as for mechanism arrows. Bond geometry (length and angles) is computed by the layout engine and is not a per-bond style; use `bond-length` to size all bonds together.

### 3.17 lone-pairs

Set `lone-pairs` to `"dots"` or `"lines"` to annotate the skeletal drawing with non-bonding electron pairs on common organic heteroatoms and charged atoms. The default `none` keeps lone pairs hidden.

```
1 #smiles("CCO", lone-pairs:  
2   "dots") \  
3 #smiles("CCN", lone-pairs: "lines") \  
4 #smiles("[O-]C=O", lone-pairs: "dots")
```

 Typst

**Note.** Lone pairs are inferred for a conservative organic subset such as N, O, S, P, halogens, and simple charged forms. They are an annotation on the skeletal structure; the carbon skeleton and implicit-hydrogen conventions do not change. For example, ammonium (`[NH4+]`) has no nitrogen lone pair. On terminal hydrogen-bearing labels such as `OH`, `OH-`, and `NH2`, lone pairs are placed from the rendered heavy-atom glyph center. For charged bare atoms such as `[O-]` or `[Br-]`, the charge mark does not move the atom center used for lone pairs.

## 4 Atoms and charges

Standard SMILES syntax (atoms, bonds, branches, ring closures) is supported. This section covers the points specific to typed-smiles.

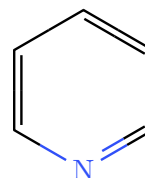
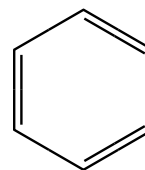
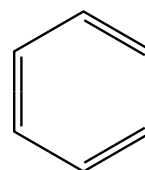
**Note.** Ring closures may be written next to branch points. For example, C1=CCCC(=O)1 closes the ring on the same atom that carries the C = O branch.

### 4.1 Aromatic rings

Aromatic rings can be written in either OpenSMILES form: lowercase aromatic atoms (c1ccccc1) or Kekulé notation with explicit alternating double bonds (C1=CC=CC=C1). Aromatic input is kekulized on parse, so both render identically.

```
1 #smiles("c1ccccc1") \  
2 #smiles("C1=CC=CC=C1") \  
3 #smiles("c1ccncc1")
```

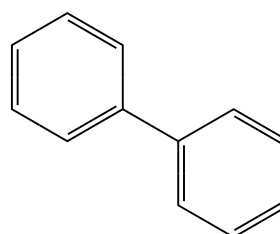
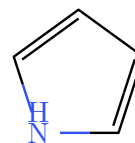
 Typst



Bracket aromatic atoms carry their hydrogen count explicitly, as in pyrrole's `[nH]`. A single bond between two aromatic rings (biphenyl) may be written with or without the explicit `-`.

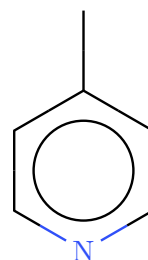
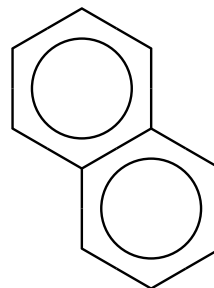
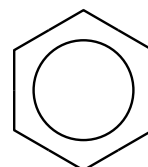
```
1 #smiles("c1cc[nH]c1") \
2 #smiles("c1occc1") \
3 #smiles("c1ccccc1-c1ccccc1", scale: 0.8)
```

 Typst



**aromatic:** "circle" depicts lowercase-aromatic rings as single bonds with an inscribed circle instead of alternating double bonds. Each fully aromatic ring of a fused system gets its own circle; Kekulé-written input always keeps its explicit bonds.

```
1 #smiles("c1ccccc1", aromatic:  
  "circle") \  
2 #smiles("c1ccc2ccccc2c1", aromatic:  
  "circle") \  
3 #smiles("Cc1ccncc1", aromatic: "circle")
```

 Typst

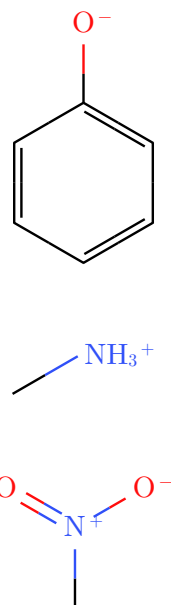
**Note.** Kekulization does not renumber anything: atom indices follow SMILES writing order for aromatic input exactly as for Kekulé input, so `show-indices`, `highlight(...)`, and `arrow(...)` references work unchanged, including on ring bonds whose double bond only appears after kekulization. A ring that cannot be kekulized (for example pyrrole written `c1ccncc1`, without the `[nH]` hydrogen) or an aromatic atom outside a ring is reported as an error.

## 4.2 Charges

Formal charges inside brackets render as a raised sign after the atom. Hydrogen counts and charge marks use the atom-label size so they remain legible when labels are scaled.

```
1 #smiles("[O-]C1=CC=CC=C1") \
2 #smiles("C[NH3+]") \
3 #smiles("C[N+](=O)[O-]")
```

 Typst



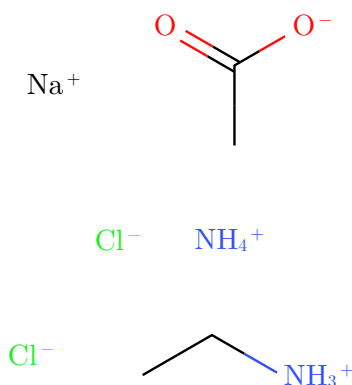
**Note.** A bracketed element such as [N] is a nitrogen atom. To print an arbitrary text label instead, use the abbreviation syntax {N} (see [Section 8](#)).

## 4.3 Salts and disconnected structures

A dot (.) means “no bond”: it separates disconnected fragments, as in salts, counterions, and hydrates. Each fragment is laid out on its own and the fragments are arranged left to right in writing order.

```
1 #smiles("CC(=O)[O-].[Na+]" ) \
2 #smiles("[NH4+].[Cl-]" ) \
3 #smiles("CC[NH3+].[Cl-]" )
```

 Typst



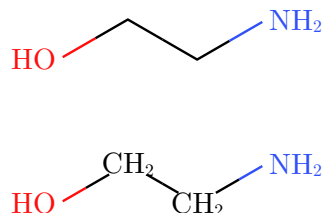
**Note.** Atom indices stay global across fragments, still following SMILES writing order, so `show-indices`, `highlight(...)`, and `arrow(...)` can reference atoms in any fragment of the same `smiles()` call. A ring closure written across a dot (as in the OpenSMILES example `C1.C1`, ethane) still forms its bond.

## 5 Hydrogens

### 5.1 Default display

Heteroatom hydrogens are shown by default; carbon hydrogens are hidden for a clean skeleton. `show-h: "all"` labels carbon hydrogens as well.

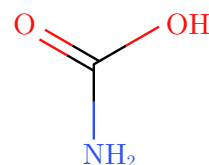
```
1 #smiles("OCCN") \
2 #smiles("OCCN", show-h: "all")
```

 Typst

### 5.2 Explicit bracket hydrogens

A hydrogen written as its own bracket atom (for example the `[H]` atoms in `C([H])([H])([H])`) is folded into its neighbor's hydrogen count, exactly like an implicit hydrogen. Carbon-bound hydrogens stay hidden and heteroatom hydrogens are still labeled, so a fully hydrogen-suppressed SMILES depicts the same as its skeletal form. Hydrogen counts written inside a single bracket, such as `[OH-]` or `[NH4+]`, are unaffected and still shown.

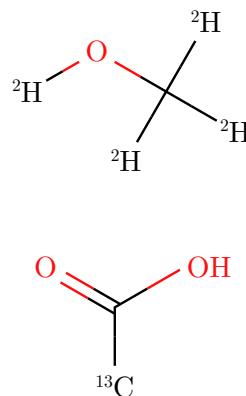
```
1 #smiles("C([H])([H])([H])[H]") \
2 #smiles("N([H])([H])C(=O)O")
```

 Typst

### 5.3 Isotopes

A bracket isotope is drawn as a leading superscript mass number. Isotopically labeled hydrogens such as deuterium `[2H]` are kept as drawn atoms rather than folded away.

```
1 #smiles("[2H]OC([2H])([2H])[2H]") \
2 #smiles("[13CH3]C(=O)O")
```

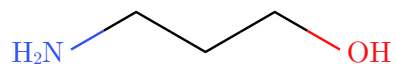
 Typst

## 5.4 Label orientation

Terminal heteroatom labels orient so the bond meets the heavy atom rather than the trailing hydrogen, at any angle.

1 `#smiles("NCCCO")`

 Typst





## 6 Stereochemistry

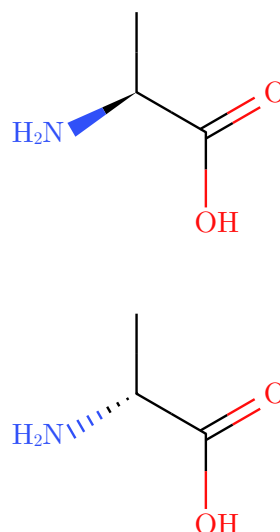
typed-smiles supports both standard SMILES stereo notations, plus a manual override for drawing wedges directly.

### 6.1 Tetrahedral centers: @ and @@

A bracket atom carrying @ or @@ becomes a wedge (toward the viewer) or hashed bond (away), computed from the 2D layout so the depiction matches the SMILES in the conventional orientation.

```
1 #smiles("N[C@@H](C)C(=O)O") \
2 #smiles("N[C@H](C)C(=O)O")
```

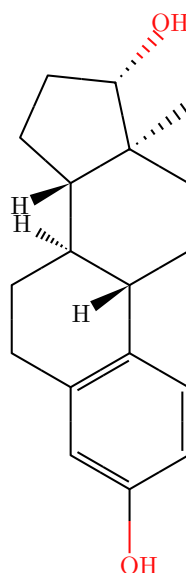
 Typst



The renderer wedges an exocyclic substituent (such as OH or CH<sub>3</sub>) where one exists, and an explicit hydrogen otherwise. Fused systems work too.

```
1 #smiles(
2   "C[C@]12CC[C@H]3[C@H]
3   ([C@@H]1CC[C@@H]2O)CCC4=C3C=CC(=C4)O",
4   scale: 0.85,
```

 Typst

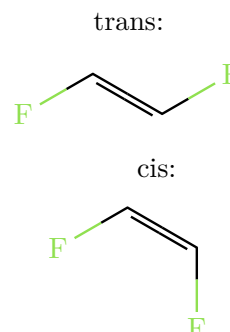


**Note.** The geometry is drawn correctly, but R/S descriptors are not computed. Ring stereochemistry between adjacent centers and bridged bicyclics may need a manual adjustment (see [Section 17](#)).

## 6.2 Double bonds: / and \

Directional bonds / and \ around a double bond set its cis/trans geometry. They come in pairs, one on each side of the =. If an alkene carbon has two marked substituent bonds, the markers must place those substituents on opposite sides.

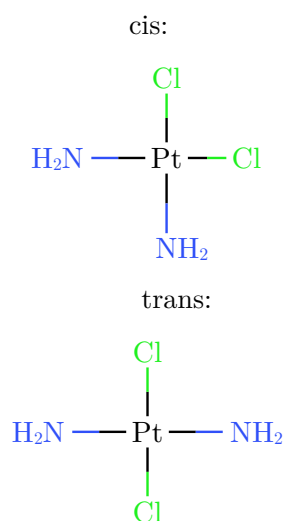
```
1 trans: #smiles("F/C=C/F")
2 #h(2em)
3 cis: #smiles("F/C=C\F")
```

 Typst

## 6.3 Square-planar centers: @SP1, @SP2, @SP3

Square-planar stereochemistry is depicted exactly: the four neighbors sit at 90° around the center, arranged so the line traced through them in SMILES order forms the class's shape — 'U' for @SP1, '4' for @SP2, 'Z' for @SP3. The classic example is cis- versus trans-platin.

```
1 cis: #smiles("N[Pt@SP1](N)
  (Cl)Cl")
2 #h(2em)
3 trans: #smiles("N[Pt@SP2](N)(Cl)Cl")
```

 Typst

**Note.** Trigonal-bipyramidal (@TB1–@TB20), octahedral (@OH1–@OH30), and allene (@AL1, @AL2) centers are accepted and drawn with correct connectivity, but without stereo decoration — their 3D arrangement is not translated into wedges.

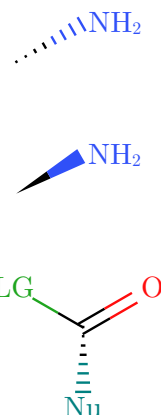
**Note.** Cumulated double bonds always label the central sp carbon (O = C = O, ketenes, allenes): the neighbors are collinear, so without the C the two double bonds would read as one long double bond.

## 6.4 Manual wedges: !w and !h

For direct control, !w draws a solid wedge and !h a hashed wedge. These are typed-smiles extensions, not standard SMILES. Like plain bonds, they are colored by the atoms at each end.

```
1 #smiles("C!wN") \
2 #smiles("C!hN") \
3 #smiles("C(!w{Nu|teal})(!LG|green)=O")
```

 Typst

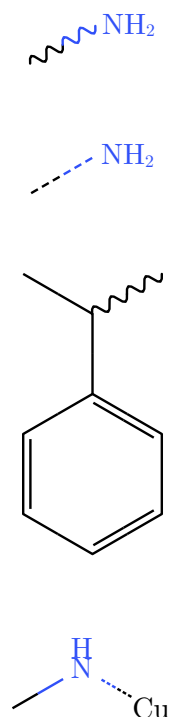


## 6.5 Wavy and dashed bonds: !s and !d

Two more drawing extensions on single bonds: !s draws a wavy (squiggly) bond — the conventional depiction of unspecified stereochemistry or an attachment point — and !d draws a dashed bond for hydrogen bonds, partial bonds, and coordination. Like plain bonds, they are colored by the atoms at each end.

```
1 #smiles("C!sN") \
2 #smiles("C!dN") \
3 #smiles("CC(!sC)C1=CC=CC=C1") \
4 #smiles("CN!d[Cu]")
```

 Typst



**Note.** !s and !d carry no cis/trans meaning and never consume a double-bond geometry slot; they can appear next to real / and \ directional bonds.

**Note.** Tip: Use empty custom labels {} or !w{} to draw bonds without a connecting atom.

## 7 Colors

### 7.1 Jmol CPK palette

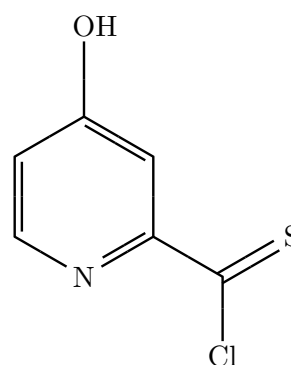
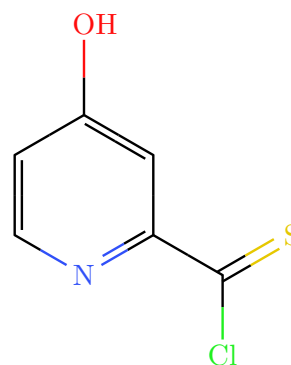
Atoms use the Jmol CPK palette by default. Carbon and unlisted elements are black; common heteroatoms get their conventional colors.

| N                                                                                 | O                                                                                 | S                                                                                 | P                                                                                 | F                                                                                 | Cl                                                                                 | Br                                                                                  | I                                                                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |  |  |  |  |

Set `color: false` for a monochrome diagram.

```
1 #smiles("OC1=CC(=NC=Cl)C(=S)Cl")
  \
2 #smiles("OC1=CC(=NC=Cl)C(=S)Cl", color:
  false)
```

 Typst



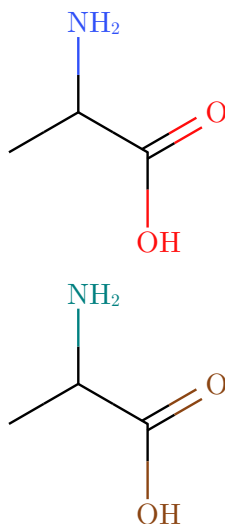
### 7.2 Custom colors (`atom-colors`)

`atom-colors` is a single dictionary that overrides colors for both elements and arbitrary abbreviated groups. Any Typst color value works — `rgb()`, `luma()`, named constants, or anything else.

## Element-symbol keys

Use an element symbol as the key to replace that element's CPK color everywhere it appears, including when referenced as an inline label style (`{OMe|O}` would pick up the overridden oxygen color).

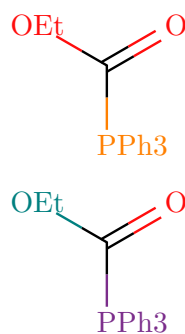
```
1 // default CPK
2 #smiles("CC(N)C(=O)O")
3 // oxygen → dark brown; nitrogen → teal
4 #smiles("CC(N)C(=O)O",
5   atom-colors: (O: rgb("#8B4513"), N: rgb("#008080")))
```

 Typst

## Label-name keys

Wrap the label text in braces to target a specific abbreviated group, regardless of any inline style it carries. The key is written as a quoted string because `{` is not a valid Typst identifier character.

```
1 // default – colors come from inline styles or element fallback
2 #smiles(">PPh3|P)C({OEt|O})=O")
3 // override by label name – inline style is ignored for these
4 #smiles(">PPh3|P)C({OEt|O})=O",
5   atom-colors: ("{PPh3}": rgb("#7B2D8B"), "{OEt}": rgb("#008080")))
```

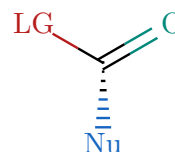
 Typst

## Mixing both key forms

Element keys and label keys can coexist in the same dictionary.

```
// 0 (element) → teal; {Nu}
1 (label) → navy; {LG} (label) → maroon
2 #smiles("C(!w{Nu})({LG|red})=O",
3   atom-colors: (0: rgb("#00897B"), "{Nu}":
   rgb("#1565C0"), "{LG}": rgb("#B71C1C")))
```

 Typst



## Priority

Color is resolved in this order for each atom or label:

1. Label-name key in `atom-colors` (e.g. "{PPh3}"), if the atom is an abbreviation.
2. Element-symbol key in `atom-colors` (e.g. 0), for both real atoms and element-style labels.
3. The inline `{label|style}` style from the SMILES string (named color, hex, or element symbol).
4. The default CPK palette.

**Note.** `color: false` is a hard override that sits above all of the above. When it is set, every atom and label renders in black regardless of any `atom-colors` entries or inline styles. If you want a mostly-monochrome diagram with one or two highlighted groups, keep or set `color: true` and put everything else in `atom-colors`.

## 7.3 Label colors ({label|style})

In the SMILES string, write `{label|style}` to color an abbreviated group independently of its element. The style can be a named color, an element symbol, or a hex code.

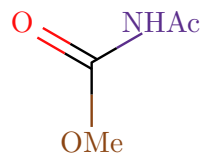
**Named colors:**

| Name   | Swatch                                                                              | Name   | Swatch                                                                              |
|--------|-------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------|
| red    |  | orange |  |
| yellow |  | brown  |  |
| green  |  | lime   |  |
| teal   |  | cyan   |  |
| blue   |  | navy   |  |
| purple |  | pink   |  |
| black  |  | gray   |  |
| silver |  | maroon |  |
| white  |  |        |                                                                                     |

**Hex colors:** write a #RRGGBB hex code after the pipe to use any color. The # is just a regular character inside a Typst string literal.

```
1 #smiles("{OMe|#8B4513}C(=O){NHAc|#5B2E8C}")
```

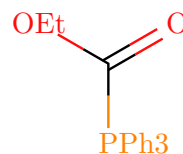
 Typst



**Element symbol:** using a symbol as the style applies that element's (possibly overridden) CPK color to the label and its bonds.

```
1 #smiles(">PPh3|P}C({OEt|O})=O")
```

 Typst



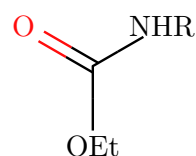
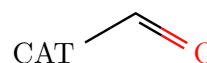
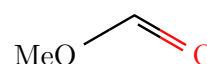
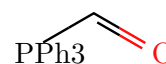


## 8 Abbreviated groups

### 8.1 Plain labels

Wrap text in braces `{...}` to place a labeled pseudo-atom that bonds like any other atom. Use `>` inside the label to choose the attachment glyph; the marker is not displayed. Rotate the molecule when needed so the bond approaches the chosen glyph roughly perpendicular to the written label.

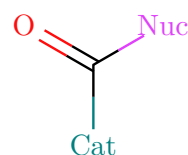
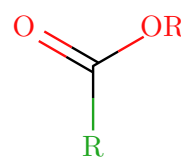
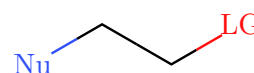
```
1 #smiles(">PPh3C=O") \
2 #smiles("{Me>O}C=O") \
3 #smiles("{CA>T}C=O") \
4 #smiles("{OEt}C(=O){NHR}")
```

 Typst

### 8.2 Colored labels

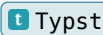
Add `|style` inside the braces to color a label. Use a named color, an element symbol, or a hex code — see [Section 7](#) for the full list.

```
1 #smiles("{Nu|blue}CC{LG|red}") \
2 #smiles("{R|green}C(=O){OR|O}") \
3 #smiles("{Cat|teal}C(=O){Nuc|#E040FB}")
```

 Typst

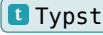
## 9 Chemical formulas: `ce()`

For formulas and text equations, `ce` is re-exported from the [chemformula](#) package, so one import covers both structures and formulas.

|                                                                                       |                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>1 #ce("H2SO4") #h(1em) #ce("Ca^2+")   #h(1em) #ce("2 H2 + O2 -&gt; 2 H2O")</pre> |  Typst $\text{H}_2\text{SO}_4 \quad \text{Ca}^{2+} \quad 2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$ |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 9.1 Fonts and size

`ce` accepts `font` and `font-size` any other arguments pass through to `chemformula`.

|                                                                                                                                                                      |                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>1 #ce("CuSO4 * 5 H2O") \ 2 #ce("CuSO4 * 5 H2O", font: "Libertinus   Serif", font-size: 13pt) \ 3 #ce("CuSO4 * 5 H2O", font: "PT Sans",   font-size: 13pt)</pre> |  Typst $\begin{array}{c} \text{CuSO}_4 \cdot 5\text{H}_2\text{O} \\ \text{CuSO}_4 \cdot 5\text{H}_2\text{O} \\ \text{CuSO}_4 \cdot 5\text{H}_2\text{O} \end{array}$ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 10 Molecular weight: `mol-weight()`

`mol-weight(smiles)` returns the molecular weight in g/mol as a plain `float`: the sum of IUPAC standard atomic weights over every atom, including implicit and explicit hydrogens. Round it for display with `calc.round`.

```
1 Ethanol: #calc.round(mol-  
weight("CCO"), digits: 2) g/mol \  
2 Caffeine: #calc.round(  
3   mol-  
weight("CN1C=NC2=C1C(=O)N(C(=O)N2C)C"),  
4   digits: 2,  
5 ) g/mol \  
6 Sodium acetate: #calc.round(  
7   mol-weight("CC(=O)[O-].[Na+]"),  
8   digits: 2,  
9 ) g/mol
```

 Typst

Ethanol: 46.07 g/mol  
Caffeine: 194.19 g/mol  
Sodium acetate: 82.03 g/mol

Dot-separated fragments are summed together, so salts and hydrates work as written. Charges are accepted; as in standard formula-weight arithmetic, the electron mass difference of ions is ignored.

**Note.** Compilation fails with a descriptive error when the weight is undefined: wildcard `*` atoms, `{label}` abbreviations (no defined composition), and isotope-labeled atoms such as `[2H]` (a specific nuclide needs a nuclide mass, not a standard atomic weight).

## 11 Inline molecules: `smiles-inline()`

`smiles-inline(smiles-str, height: 1.4em, baseline: auto, ..args)` scales a structure to a target height and baseline-aligns it so it reads inline, like a word in the sentence. Extra named arguments are passed straight to `smiles()`.

```
1 Dehydrating ethanol #smiles-  
  inline("CCO")  
2 over alumina gives ethylene  
3 #smiles-inline("C=C"); toluene  
4 #smiles-inline("Cc1ccccc1") is a common  
5 solvent.
```

 Typst

Dehydrating ethanol  over alumina  
gives ethylene  toluene  is a common  
solvent.

Sizing works in two steps: the drawing is scaled to fit the `height` (default `1.4em`, so ordinary line spacing stays essentially unchanged), and the scale is capped so one bond never exceeds most of that height — otherwise a flat, mostly horizontal molecule, which measures almost no height, would blow up to fill it. Typst grows a line to fit tall inline content, so neighboring lines are never overlapped; passing a larger `height` only makes the molecule's own line taller.

**Note.** `baseline` sets how far the drawing's vertical center sits above the text baseline (default centers it on the lowercase body). Atom labels scale down with the drawing: at inline sizes, label-heavy molecules read better with a larger `height` or as a display-mode `smiles()`.

## 12 CeTZ integration: smiles-cetz()

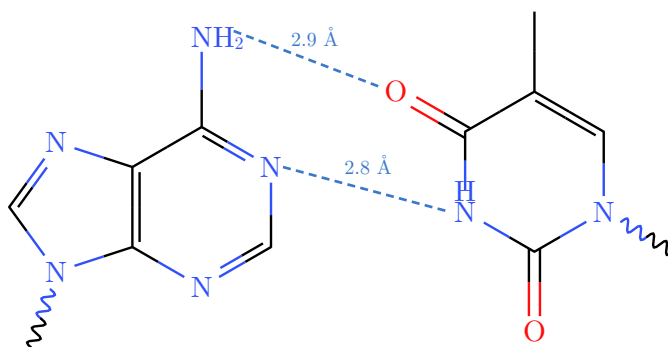
`smiles-cetz(smiles-str, name:, origin:, fg:, theme:, ..opts)` draws a molecule **inside an existing CeTZ canvas** and registers named anchors on the group, so arbitrary CeTZ drawing attaches to real molecular positions:

- "`<name>.atom-<i>`" — every atom center (writing-order indices, as shown by `show-indices`),
- "`<name>.bond-<i>-<j>`" — every bond midpoint (with  $i < j$ ),
- "`<name>.center`" — the molecule origin.

Coordinates are in bond-length units; give the canvas `length: 30pt * scale` so sizes match `smiles(scale: ...)`, and wrap the canvas in `context` (atom-label measurement needs it — plain `smiles()` provides it automatically, a raw canvas does not).

The Watson–Crick A–T base pair below is two `smiles-cetz` calls plus plain CeTZ: two dashed hydrogen bonds drawn between atom anchors of the two bases, endpoint nudges so the dashed lines do not start at atom centers, distance labels, and glycosidic-bond stubs to the sugar backbone.

```
1  #align(center, context cetz.canvas(length: 30pt, {
2  import cetz.draw: *
3
4  smiles-cetz("Nc1ncnc2N(!s{ })cnc12", name: "A")
5  smiles-cetz("Cc1cN(!s{ })c(=O)[nH]c1=O", name: "T", origin: (4.9, 0.42))
6
7  let hb = (paint: rgb("#3A78C9"), thickness: 1.0pt, dash: "densely-dashed")
8  let off(anchor, by) = (rel: by, to: anchor)
9  line(off("A.atom-11", (0.4, -0.15)), off("T.atom-9", (-0.2, 0.06)), stroke: hb)
10 line(off("A.atom-2", (0.15, 0)), off("T.atom-7", (-0.2, 0)), stroke: hb)
11
12 // Hydrogen-bond distances.
13 content((rel: (0.2, 0.2), to: ("A.atom-11", 50%, "T.atom-9")), text(size: 7.5pt, fill:
14   rgb("#3A78C9"))[2.9 Å])
15 content((rel: (0, 0.28), to: ("A.atom-2", 50%, "T.atom-7")), text(size: 7.5pt, fill:
16   rgb("#3A78C9"))[2.8 Å])
17 })))
```

**t** Typst

The remaining `..opts` are `smiles()` drawing options (color, rotation, mirror, show-h, aromatic, opacity, bond-customizations, ...). `fg` and `theme` default to black and light because auto foreground resolution is not available inside a raw canvas — pass them explicitly on dark backgrounds.

Use regular CeTZ relative coordinates to offset any connection endpoint: (`rel: (0.12, 0.04)`, `to: "A.atom-0"`). This is the same idea as `atom(..., offset:)` in mechanism arrows, but in raw CeTZ syntax.

**Note.** Turn on `show-indices: true` while writing anchor references, exactly as for mechanism arrows, to read off the atom indices. Bond-midpoint anchors (`bond-<i>-<j>`) attach to the middle of a bond — useful for marking a reacting or attachment bond.

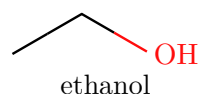
## 13 Reaction schemes

Three helpers compose molecules, formulas, and arrows into schemes.

### 13.1 mol()

`mol(spec, label: none, offset: (0,0), ..opts)` is a reaction item. `spec` is either any content (`smiles(...)`, `ce(...)`, `text`) or a SMILES **string**. Passing a string lets `reaction` render the molecule itself so its atoms become addressable by curly arrows (see [Reaction mechanisms](#)). String molecules accept common drawing options such as `font-size`, `font`, `bond-stroke`, `color`, `rotation`, `show-h`, `lone-pairs`, `opacity`, `bond-customizations`, `atom-colors`, and `show-indices`. `offset` nudges the molecule in page coordinates, in bond-length units: positive `x` moves right and positive `y` moves up regardless of `reaction(flow:)`. In mechanism mode, use `reaction(scale: ...)` to resize the shared canvas and its offsets.

```
1 #reaction(mol(smiles("CCO"),  
  label: [ethanol]))
```

 Typst

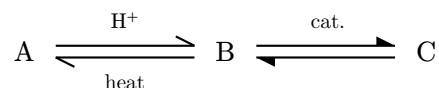
### 13.2 rxn-arrow()

`rxn-arrow(above: none, below: none, dir: auto, kind: "single")` draws an arrow. `dir` may be "right", "left", "up", "down", or `auto` (the default, following the enclosing `reaction(flow:)`). `kind` may be "single", "equilibrium", "equilibrium-filled", "dashed", or "wavy". `above` and `below` carry reagents and conditions. `color` defaults to `auto`, inheriting the surrounding text color (so arrows recolor on dark slides).

| Argument | Default  | Effect                                                                                                        |
|----------|----------|---------------------------------------------------------------------------------------------------------------|
| above    | none     | Label above a horizontal arrow, or right of a vertical arrow.                                                 |
| below    | none     | Label below a horizontal arrow, or left of a vertical arrow.                                                  |
| dir      | auto     | Arrow direction: "right", "left", "up", "down", or <code>auto</code> (follows <code>reaction(flow:)</code> ). |
| kind     | "single" | Arrow style: "single", "equilibrium", "equilibrium-filled", "dashed", or "wavy".                              |
| color    | auto     | Arrow color; <code>auto</code> inherits the surrounding text color.                                           |

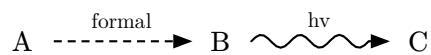
Equilibrium arrows use paired half-heads. The filled variant uses filled half-heads.

```
1 #reaction(  
2   ce("A"),  
3   rxn-arrow(kind: "equilibrium", above: ce("H+"), below: [heat]),  
4   ce("B"),  
5   rxn-arrow(kind: "equilibrium-filled", above: [cat.]),  
6   ce("C"),  
7 )
```

[t Typst](#)

A dashed arrow marks a hypothetical or formal transformation; a wavy arrow marks a distorted or non-elementary one.

```
1 #reaction(  
2   ce("A"),  
3   rxn-arrow(kind: "dashed", above: [formal]),  
4   ce("B"),  
5   rxn-arrow(kind: "wavy", above: [hv]),  
6   ce("C"),  
7 )
```

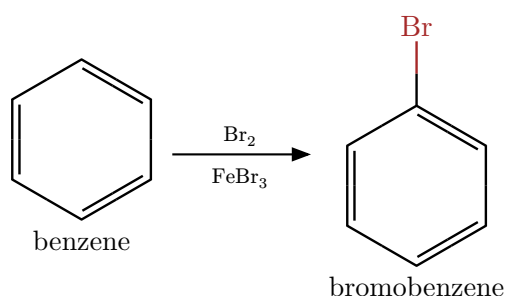
[t Typst](#)



### 13.3 reaction()

`reaction(gap-h: 1.5em, gap-v: 1.5em, scale: 1.0, breakable: false, show-indices: false, flow: "right", ..items)` lays out molecules and arrows left to right. An up or down arrow wraps the scheme onto a new row.

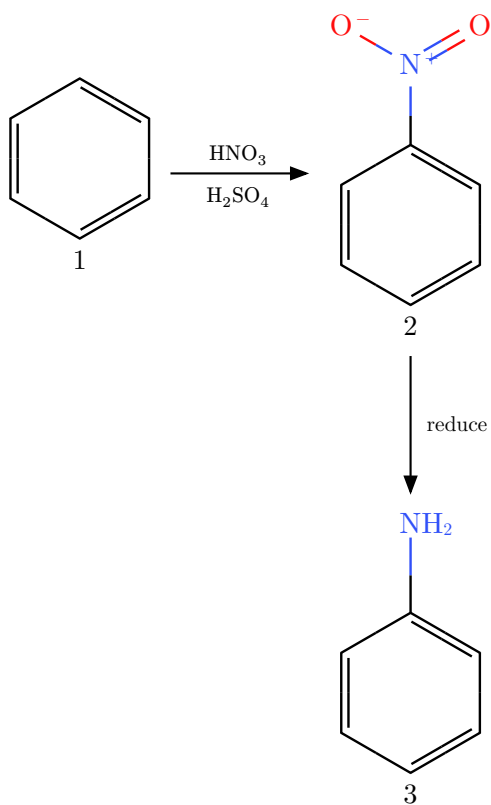
```
1 #reaction(  
2   mol(smiles("C1=CC=CC=C1"), label: [benzene]),  
3   rxn-arrow(above: ce("Br2"), below: ce("FeBr3")),  
4   mol(smiles("BrC1=CC=CC=C1"), label: [bromobenzene]),  
5 )
```

 Typst

**Note.** Set `reaction(show-indices: true)` to stamp indices on every string SMILES molecule rendered through `mol("...")` in that reaction. This is useful while authoring large mechanisms with many arrows or highlights. A molecule can still opt out with `mol("...", show-indices: false)`. Molecules already pre-rendered as content, such as `mol(smiles("..."))`, keep their own `smiles()` options.

A wrap-around scheme using a downward arrow:

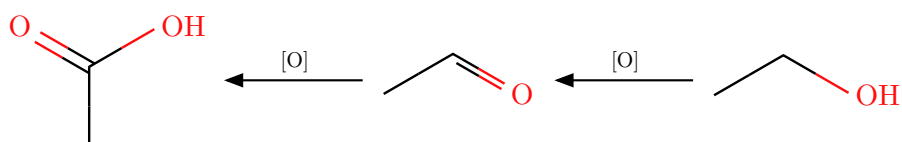
```
1 #reaction(  
2   mol(smiles("C1=CC=CC=C1"), label: [1]),  
3   rxn-arrow(above: ce("HNO3"), below: ce("H2SO4")),  
4   mol(smiles("[O-][N+](=O)C1=CC=CC=C1"), label: [2]),  
5   rxn-arrow(dir: "down", above: [reduce]),  
6   mol(smiles("NC1=CC=CC=C1"), label: [3]),  
7 )
```

[t Typst](#)

## Flow direction

`flow` sets the writing direction: "right" (default), "left", "up", or "down". "left" and "up" reflect the scheme along the flow axis and flip the arrowheads, so a scheme written in reading order displays right-to-left or bottom-to-top. Ordinary non-arrow items follow the same direction, so `flow: "up"` and "down" stack molecules, plus signs, and text vertically even when no `rxn-arrow()` separators are present. This is the natural choice for a branch that grows out of the left or bottom of a `cycle()` — its released product ends up on the side facing the cycle. Arrows created with `rxn-arrow(dir: auto)` (the default) follow the flow; an explicit `dir` is kept.

```
1 #reaction(flow: "left",
2   mol(smiles("CCO")),
3   rxn-arrow(above: [\[O\]]),
4   mol(smiles("CC=O")),
5   rxn-arrow(above: [\[O\]]),
6   mol(smiles("CC(=O)O")),
7 )
```

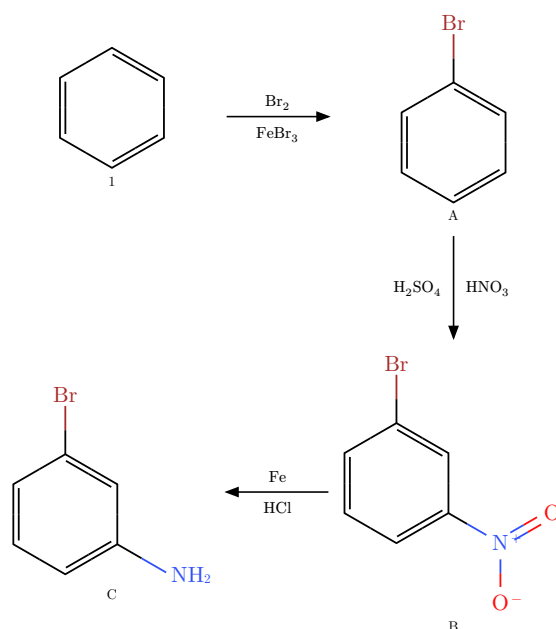
 Typst

**Note.** Because `step(out:)` of a `cycle()` accepts any content, a nested `reaction(flow: "left")` (or "down") grows a full sub-reaction out of a released molecule that reads naturally away from the ring, while the outer reaction continues in its own direction. Flow applies to the scheme layout; mechanism mode (curly arrows) is unaffected.

## Uniform scale

Pass `scale` to shrink or enlarge the entire scheme uniformly — molecules, arrows, labels, and separators all change together. This is especially useful for multi-step or wrap-around schemes that would otherwise be too wide.

```
1 // same scheme at three different scales
2 #reaction(
3   scale: 0.75,
4   mol(smiles("C1=CC=CC=C1"), label: text(size: 7pt)[1]),
5   rxn-arrow(above: ce("Br2"), below: ce("FeBr3")),
6   mol(smiles("BrC1=CC=CC=C1"), label: text(size: 7pt)[A]),
7   rxn-arrow(dir: "down", above: ce("HNO3"), below: ce("H2SO4")),
8   mol(smiles("BrC1=CC(=CC=C1)[N+](=O)[O-]"), label: text(size: 7pt)[B]),
9   rxn-arrow(dir: "left", above: ce("Fe"), below: ce("HCl")),
10  mol(smiles("BrC1=CC(=CC=C1)N"), label: text(size: 7pt)[C]),
11 )
```

 Typst

**Note.** `reaction(scale: ...)` and `smiles(scale: ...)` are independent. The reaction scale is applied on top of however each individual molecule is sized. To resize everything from a single place, use `reaction(scale: ...)` and let each `smiles()` call use its default.

## Page-break behaviour

By default, `reaction` sets `breakable: false`, so the whole scheme moves to the next page as a unit if it does not fit on the current one. This prevents a molecule or vertical branch from stranding on a different page from the rest of the scheme. Set `breakable: true` only for very long schemes that must span pages.

```
1 // Default – the scheme always stays on one page:
2 #reaction( ... )           // breakable: false
3
4 // Opt in to page splitting for very long schemes:
5 #reaction(breakable: true, ... )
```



## 14 Reaction mechanisms

`reaction()` also draws electron-pushing mechanisms: curly arrows that flow from a lone pair, bond, or atom into another bond or atom — within one structure or across separate species. It switches from grid layout to a single shared canvas as soon as any curly `arrow()` or `highlight()` is present. `mol(offset:)` by itself stays in scheme layout and nudges the species inside its grid cell.

### 14.1 Referencing atoms

Atoms are addressed by their **writing-order index** (0-based), so the SMILES string is never modified. Inside `smiles()` use single-index references; inside `reaction()` prefix them with the species index `s` — the position of the `mol()/content` item in written order (`rxn-arrows` and annotations are not counted).

| Reference                                            | Resolves to                                                                |
|------------------------------------------------------|----------------------------------------------------------------------------|
| <code>atom(i)</code> / <code>atom(s, i)</code>       | Atom center.                                                               |
| <code>bond(i, j)</code> / <code>bond(s, i, j)</code> | Midpoint of the bond between atoms <code>i</code> and <code>j</code> .     |
| <code>lp(i)</code> / <code>lp(s, i)</code>           | A lone pair on atom <code>i</code> (use <code>pair: n</code> to pick one). |
| <code>species(k)</code>                              | Bounding-box edge of a whole item (e.g. a <code>ce()</code> formula).      |

**Note.** Every reference accepts an `offset: (dx, dy)` nudge in bond-length units — the escape hatch for fine-tuning an endpoint or pointing at a single-electron dot. Set `show-indices: true` on a `smiles()` / `mol()` to stamp the indices on the diagram while you write the arrows.

Hydrogen atoms declared inside bracket notation (e.g. `[OH-]`, `[NH4+]`) are also addressable. Each H gets the next available index after all heavy atoms, so `atom(1)` in `[OH-]` refers to the hydrogen. With `show-indices`, the heavy-atom and H badges are centered on their rendered label glyphs. Charge marks are excluded from the atom center, so `atom(0)` in `[O-]` points to the O glyph rather than the combined `O-` label.

## 14.2 `arrow()` and `highlight()`

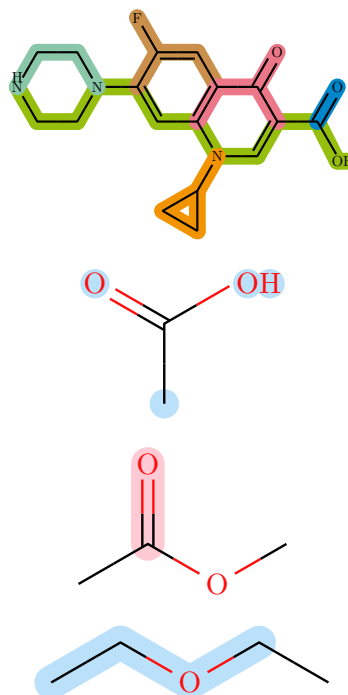
`arrow(from:, to:, label: none, color: red, bend: "left", angle: 35deg, half: false)` draws a curly arrow between two references. `bend` is "left", "right", or none `angle` sets how strongly it bows; `half: true` draws a fishhook (single-electron) head. `highlight(ref, fill:)` shades an atom (disk) or bond (capsule) behind the structure. Pass an array of references to shade several atoms or bonds with one call. Bond highlights are trimmed away from atom centers by default; set `include-atoms: true` to also shade the endpoint atoms of each highlighted bond. Abbreviated group labels such as {PPh3} are highlighted as measured label-width capsules rather than atom-sized disks.

```

1  #smiles(
2      "N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)
    highlight((bond(0, 5), bond(5, 4),
    bond(4, 3), bond(3, 6), bond(6, 10),
    bond(10, 11), bond(11, 15), bond(15,
3      19), bond(19, 20), bond(20, 21),
    bond(21, 23), bond(23, 25)), fill:
    rgb(150, 191, 13), include-atoms:
    true),
    highlight((bond(15, 16), bond(16, 18),
4      bond(18, 17), bond(17, 16)), fill:
    rgb(242, 148, 1), include-atoms: true),
    highlight((bond(3, 2), bond(2, 1),
5      bond(1, 0)), fill: rgb(137, 199, 168),
    include-atoms: true),
    highlight((bond(6, 7), bond(7, 8),
6      bond(7, 9), bond(9, 12)), fill:
    rgb(201, 143, 75), include-atoms:
    true),
    highlight((bond(11, 12), bond(12, 13),
7      bond(13, 20), bond(13, 14)), fill:
    rgb(236, 119, 137), include-atoms:
    true),
8      highlight((bond(21, 22)), fill: rgb(0,
    134, 203), include-atoms: true),
9      color: false,
10     rotation: 90deg,
11     bond-stroke: 0.8pt,
12     scale: 0.5,
13 )
14 #smiles(
15     "CC(=O)O",
16     highlight((atom(0), atom(2), atom(3),
    atom(4)), fill: rgb("#BBE1FA")),
17 )
18 #smiles(
19     "CC(=O)OC",
20     highlight((bond(2, 1)), include-
    atoms:true, fill: rgb("#FFCAD4")),
21 )
22 #smiles(
23     "CCOCC",
24     highlight(
25         (bond(0, 1), bond(1, 2), bond(2, 3)),
26         fill: rgb("#BBE1FA"),
27         include-atoms: true,
28     ),
29 )

```

 Typst



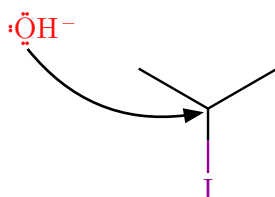


### 14.3 A mechanism across species

Hydroxide attacks the central carbon; the C-I bond breaks toward iodide. The nucleophile is offset so the curly arrow reads cleanly.

```
1 #reaction(
2   mol("[OH-]", lone-pairs: "dots", offset: (0, 1)),
3   mol("C(I)(C)C"),
4   arrow(from: lp(0, 0, offset:(0.1, -0.2)), to: atom(1, 0, offset : (0.1, -0.1)),
5         bend: "right", color : black),
6 )
```

 Typst

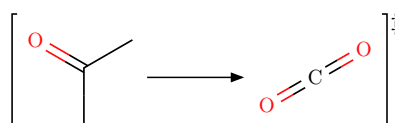


### 14.4 brackets()

`brackets(body, sup: none, sub: none)` encloses any content in square brackets including another reaction, with optional marks at the corners — a `[‡]` for a transition state or a charge for a reactive intermediate.

```
1 #brackets(
2   [#reaction(smiles("CC(=O)C"), rxn-
3     arrow(), smiles("O=C=O"), scale : 0.7)],
4   sup: [‡])
```

 Typst



## 15 Catalytic cycles: `cycle()`

---

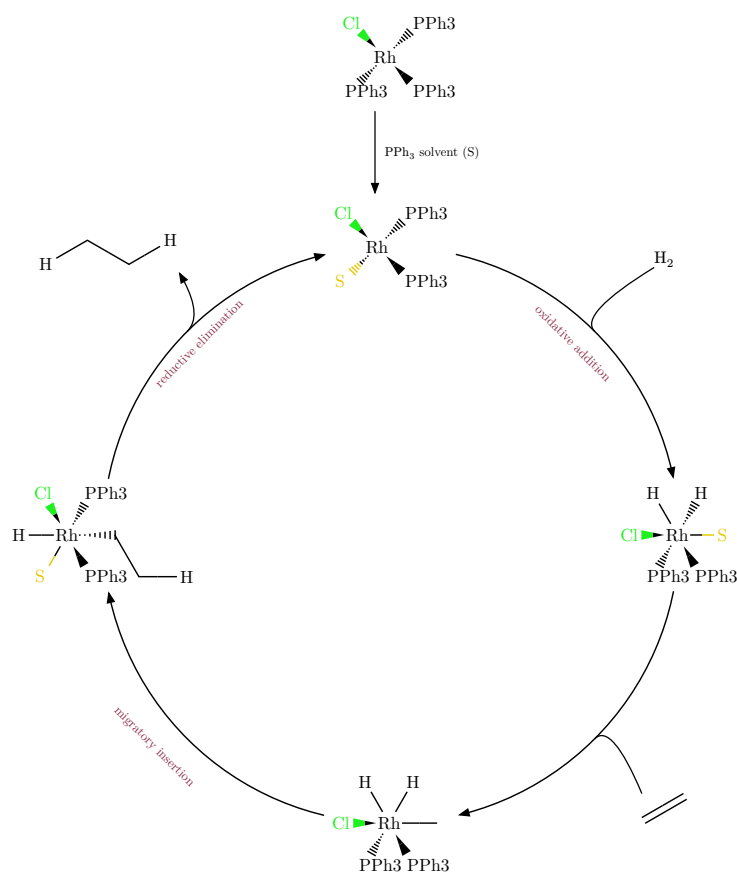
`cycle(..items, radius:, start:, clockwise:, scale:, reagent-bend:, ...)` arranges species on a circle with arc arrows between them — the standard depiction of a catalytic cycle. Items alternate species and `step()`s, just as `reaction()` alternates molecules and arrows, but the sequence closes into a ring (the last step returns to the first species). Species are `mol()` items or any content; SMILES strings are rendered by `smiles()`.

`step(label:, into:, out:, bend:, merge:, rotation:, label-offset:, into-offset:, out-offset:)` describes the arc that follows a species: `label` names the transformation, `into` adds a reagent merging into the arc from outside the ring, and `out` a product leaving it. Those side arrows are bookkeeping arrows for material entering or leaving the catalytic manifold, not electron-pushing arrows. `merge: true` draws a side arrow tangent to the arc so it visually fuses with the main cycle arrow. `rotation` rotates the step label: "straight" keeps it horizontal, "auto" follows the step's circle angle while keeping text upright, and an explicit angle is used as given. The three `*-offset` arguments nudge the label and the `into/out` reagents like a `mol` offset.

```

1  #align(center)[
2  #reaction(
3    scale : 0.6,
4    flow : "down",
5    mol(smiles("{Rh}(!w{>PPh3})(!h{>PPh3})(!hCl)(!w{>PPh3})"), offset: (0.5, 0)),
6    rxn-arrow(above : [#ce("PPh3") solvent (S)]),
7    cycle(
8      radius : 6.0,
9      mol(smiles("{Rh}(!w{S | S})(!h{>PPh3})(!hCl)(!w{>PPh3})"),
10     step(label : [#text(size : 8pt)[oxidative addition]], label-offset : (0.3, 0.5),
11     into : [#ce("H2")], rotation : "auto", merge : true),
12     mol(smiles("{Rh}(!w{>PPh3})(!h{>PPh3})(!hCl)({S | S})(!w{H})(!w{H})"),
13     step(into : [#smiles("C=C")], merge : true),
14     mol(smiles("{Rh}(!w{>PPh3})(!h{>PPh3})(!hCl)({})(!w{H})(!w{H})"),
15     step(label : [#text(size : 8pt)[migratory insertion]], label-offset: (0, 0.5),
16     rotation : "auto"),
17     mol(smiles("{Rh}(!w{>PPh3})(!h{>PPh3})(!hCl)({S | S})(!w{CC{H}})", rotation :
18     -90deg)),
19     step(label : [#text(size : 8pt)[reductive elimination]], label-offset: (-0.5,
20     0.5), out : [#smiles("{H}CC{H}")], merge:true, rotation : "auto")
21   )
22 )
23 ]

```

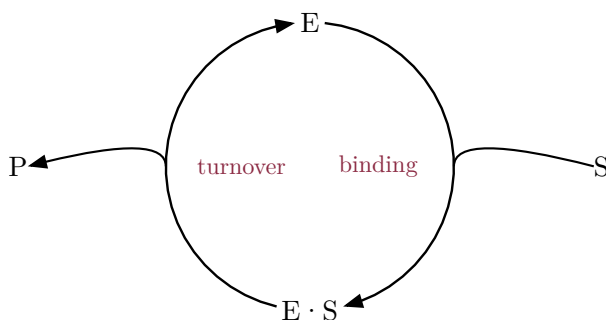


Because `step(out:)` accepts any content, passing a `reaction()` there grows a full linear branch out of a released molecule — the released product becomes the first species of a downstream sequence. Reagents (both `into` and `out`) are offset by their own measured size, so even a wide nested reaction clears the ring. Incoming content is attached on the side facing the cycle so the side arrow starts at the last upstream item; outgoing content uses the same side so the leaving arrow lands at the first downstream item instead of the center of the whole branch.

**Note.** `radius` defaults to `auto`, which fits the species without overlap; set it explicitly for a tighter or looser ring. `start` is the angle of the first species (`90deg` is the top) and `clockwise` the travel direction. `scale` sets the canvas bond length as in `reaction()`. `arc-gap` tunes how close the arc arrows sit to the species (in bond-length units beyond each molecule's radius; smaller or even negative brings them closer). `reagent-bend` sets the default curvature of into/out side arrows; `step(bend:)` overrides one step, and `bend: 0` makes that side arrow nearly radial. A one-species `cycle()` draws that species alone.

With `merge: true` the into/out arrows fuse tangentially with the main cycle arrows, and `arc-gap` pulls the arrows closer to the species:

```
1 #cycle(  
2   scale: 0.9,  
3   arc-gap: 0.0,  
4   mol(box(inset: 2pt)[E]),  
5   step(label: [binding], into: [S], merge: true),  
6   mol(box(inset: 2pt)[E·S]),  
7   step(label: [turnover], out: [P], merge: true),  
8 )
```

 Typst

## 16 Project-wide defaults

Typst functions support partial application with `.with()`. Calling `smiles.with(...)` returns a new function with the given arguments pre-filled. Rebind the name in your preamble and every subsequent `#smiles(...)` call uses those defaults — including inside `#reaction()`, because the content is evaluated before `reaction` sees it.

Any argument can be pre-filled this way: `bond-length`, `font`, `scale`, `font-size`, `atom-colors`, or any other `smiles` parameter.

```
1 // — preamble —————
2 #import "@preview/typed-smiles:0.5.0": *
3
4 #let smiles = smiles.with(
5   bond-length: 0.9,
6   font:       "Libertinus Serif",
7   atom-colors: (O: rgb("#8B4513"), N: rgb("#008080")),
8 )
9
10 // — anywhere in the document —————
11 #smiles("CC(N)C(=O)O") // uses bond-length 0.9, Libertinus, custom O/N
12 #reaction(
13   smiles("CCO"),
14   rxn-arrow(),
15   smiles("CC(=O)O"), // same custom smiles – preamble applies here too
16 )
```

 Typst

**Note.** Any argument passed directly at the call site still overrides the `.with()` default for that call alone.

## 17 Limitations

---

- Directional bonds (*/*, *\*) draw the correct cis/trans geometry, but E/Z descriptors are not computed.
- R/S and E/Z descriptors are not calculated.
- Trigonal-bipyramidal (**@TB**), octahedral (**@OH**), and allene (**@AL**) centers are accepted but drawn without stereo wedges.
- Ring stereochemistry between adjacent centers and bridged bicyclics can overlap or need a manual adjustment (try **rotation**, or the manual **!w** and **!h** wedges). Ordinary substituent branches resolve crowding automatically — a blocking branch flips to the other side of its attachment bond (e.g. the ortho acetyl and carboxyl groups of aspirin), or the colliding bond flips its zigzag turn — always keeping ideal bond angles, so plain branches do not overlap.

## 18 Quick reference

### 18.1 Syntax

| Syntax                     | Meaning                                               |
|----------------------------|-------------------------------------------------------|
| <code>c n o ...</code>     | Aromatic atom (lowercase); kekulized on parse.        |
| <code>.</code>             | Disconnected fragment (no bond): salts, hydrates.     |
| <code>[...]</code>         | Bracket atom (any element, charges, explicit H).      |
| <code>@ / @@</code>        | Tetrahedral anticlockwise / clockwise.                |
| <code>@SP1-@SP3</code>     | Square planar (U / 4 / Z shape), depicted exactly.    |
| <code>@TB @OH @AL</code>   | Extended stereo: accepted, drawn without wedges.      |
| <code>\$</code>            | Quadruple bond (four parallel lines).                 |
| <code>/ \</code>           | Double-bond cis/trans geometry.                       |
| <code>!w / !h</code>       | Manual solid / hashed wedge (typed-smiles extension). |
| <code>!s / !d</code>       | Wavy / dashed bond (typed-smiles extension).          |
| <code>{label}</code>       | Abbreviated group pseudo-atom.                        |
| <code>&gt;{label}</code>   | Abbreviated group anchored at the glyph after >.      |
| <code>{label style}</code> | Colored abbreviated group.                            |

### 18.2 smiles() options

| Option                   | Default                            | Effect                                                                                                   |
|--------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------|
| <code>style</code>       | <code>"default"</code>             | Journal preset: <code>"acs"</code> , <code>"rsc"</code> , <code>"nature"</code> , <code>"wiley"</code> . |
| <code>scale</code>       | <code>1.0</code>                   | Balanced size of everything.                                                                             |
| <code>bond-length</code> | <code>none</code>                  | Bond length only ( <code>1.0</code> is 30 pt).                                                           |
| <code>font-size</code>   | <code>none</code>                  | Atom-label size only.                                                                                    |
| <code>font</code>        | <code>"New Computer Modern"</code> | Atom-label font.                                                                                         |
| <code>bond-stroke</code> | <code>none</code>                  | Bond width only.                                                                                         |
| <code>color</code>       | <code>auto</code>                  | CPK colors for <code>"default"</code> monochrome for journal presets.                                    |
| <code>fg</code>          | <code>auto</code>                  | Foreground for bonds/carbon labels; <code>auto</code> inherits the text color.                           |
| <code>theme</code>       | <code>auto</code>                  | CPK palette variant; <code>auto</code> goes dark when <code>fg</code> is light.                          |
| <code>rotation</code>    | <code>0deg</code>                  | Rotate, labels stay upright.                                                                             |
| <code>mirror</code>      | <code>none</code>                  | Optional <code>"horizontal"</code> or <code>"vertical"</code> reflection.                                |
| <code>show-h</code>      | <code>()</code>                    | Label selected implicit hydrogens; use <code>"all"</code> for every atom.                                |

|                     |          |                                                                                |
|---------------------|----------|--------------------------------------------------------------------------------|
| aromatic            | "kekule" | Lowercase-aromatic rings as doubles or "circle".                               |
| atom-annotations    | ()       | Small gray side labels as (index, content) or (index, content, offset) tuples. |
| opacity             | 100%     | Fade the whole drawing (ghost molecules).                                      |
| bond-customizations | ()       | Per-bond color, stroke, opacity keyed by bond(i, j).                           |
| lone-pairs          | none     | Draw lone pairs as "dots" or "lines".                                          |
| atom-colors         | (:)      | Color overrides: 0: red for elements, "{PPh3}": blue for labels.               |
| show-indices        | false    | Stamp atom indices for writing references.                                     |
| ..annotations       | —        | arrow() / highlight() items on this molecule.                                  |

### 18.3 reaction() options

| Option       | Default | Effect                                                                            |
|--------------|---------|-----------------------------------------------------------------------------------|
| gap-h        | 1.5em   | Horizontal gap between items.                                                     |
| gap-v        | 1.5em   | Vertical gap between rows.                                                        |
| scale        | 1.0     | Uniform scale applied to the whole scheme.                                        |
| breakable    | false   | Allow the scheme to split across pages.                                           |
| flow         | "right" | Writing direction: "right", "left", "up", or "down" (left/up reflect the scheme). |
| show-indices | false   | Default atom-index overlay for string SMILES molecules in this reaction.          |

### 18.4 rxn-arrow() options

| Option | Default  | Effect                                                                             |
|--------|----------|------------------------------------------------------------------------------------|
| above  | none     | Condition label above a horizontal arrow, or right of a vertical arrow.            |
| below  | none     | Condition label below a horizontal arrow, or left of a vertical arrow.             |
| dir    | auto     | Arrow direction: "right", "left", "up", "down", or auto (follows reaction(flow:)). |
| kind   | "single" | Arrow style: "single", "equilibrium", "equilibrium-filled", "dashed", or "wavy".   |
| color  | auto     | Arrow color; auto inherits the surrounding text color.                             |

### 18.5 Data helpers

| Helper             | Purpose                                                                                |
|--------------------|----------------------------------------------------------------------------------------|
| mol-weight(smiles) | Molecular weight in g/mol as a float errors on wildcards, abbreviations, and isotopes. |



|                                                                       |                                                                                                                                   |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>smiles-inline(smiles, height:, baseline:, ..args)</code>        | Molecule scaled and baseline-aligned for running text.                                                                            |
| <code>smiles-cetz(smiles, name:, origin:, fg:, theme:, ..opts)</code> | Molecule as CeTZ elements with <code>atom-&lt;i&gt;</code> / <code>bond-&lt;i&gt;-&lt;j&gt;</code> / <code>center</code> anchors. |

## 18.6 Mechanism helpers

| Helper                                                                                    | Purpose                                                                                                                          |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <code>mol(spec, label:, offset:, ..opts)</code>                                           | Reaction item; <code>spec</code> as a string is rendered with addressable atoms; <code>offset</code> nudges in page coordinates. |
| <code>cycle(..items, radius:, start:, clockwise:, scale:, reagent-bend:, arc-gap:)</code> | Catalytic cycle: species on a ring with arc arrows.                                                                              |
| <code>step(label:, into:, out:, bend:, merge:, rotation:, *-offset:)</code>               | A cycle arc: transformation label, reagent in/out, side-arrow bend, tangential merge, label rotation, and per-piece offsets.     |
| <code>atom / bond / lp / species</code>                                                   | Atom-index references (optional <code>offset:</code> ).                                                                          |
| <code>arrow(from:, to:, label:, color:, bend:, angle:, half:)</code>                      | Curly electron-pushing arrow.                                                                                                    |
| <code>highlight(ref, fill:, stroke:, radius:, include-atoms:)</code>                      | Shade one atom/bond reference or                                                                                                 |

|                                         |                                             |
|-----------------------------------------|---------------------------------------------|
|                                         | an array of references.                     |
| <code>brackets(body, sup:, sub:)</code> | Square brackets with optional corner marks. |

18.7 Label color names

Named styles for {label|style} — any of these plus any #RRGGBB hex:

| Name   | Swatch                                                                            | Name   | Swatch                                                                            | Name   | Swatch                                                                            |
|--------|-----------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------|
| red    |  | orange |  | yellow |  |
| brown  |  | green  |  | lime   |  |
| teal   |  | cyan   |  | blue   |  |
| navy   |  | purple |  | pink   |  |
| black  |  | gray   |  | silver |  |
| maroon |  | white  |  |        |                                                                                   |